

```
In [37]: import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
```

```
In [38]: df = pd.read_csv(r"C:\Users\USER\Downloads\drugs.csv")
```

Load Data and understand the Dataset:

The dataset is loaded using Pandas and examined to understand its structure, number of observations, variables, data types, and missing values.

```
In [6]: df.head()
```

Out[6]:

	State	Year	Population.12-17	Population.18-25	Population.26+	Totals.Alcohol.Use Disorder Past Year.12-17
0	Alabama	2002	380805	499453	2812905	18
1	Alaska	2002	69400	62791	368460	4
2	Arizona	2002	485521	602265	3329482	36
3	Arkansas	2002	232986	302029	1687337	14
4	California	2002	3140739	3919577	21392421	173

5 rows × 53 columns



```
In [7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 867 entries, 0 to 866
```

```
Data columns (total 53 columns):
```

#	Column	Non-Null Count	Dtype
0	State	867 non-null	object
1	Year	867 non-null	int64
2	Population.12-17	867 non-null	int64
3	Population.18-25	867 non-null	int64
4	Population.26+	867 non-null	int64
5	Totals.Alcohol.Use Disorder Past Year.12-17	867 non-null	int64
6	Totals.Alcohol.Use Disorder Past Year.18-25	867 non-null	int64
7	Totals.Alcohol.Use Disorder Past Year.26+	867 non-null	int64
8	Rates.Alcohol.Use Disorder Past Year.12-17	867 non-null	float64
9	Rates.Alcohol.Use Disorder Past Year.18-25	867 non-null	float64
10	Rates.Alcohol.Use Disorder Past Year.26+	867 non-null	float64
11	Totals.Alcohol.Use Past Month.12-17	867 non-null	int64
12	Totals.Alcohol.Use Past Month.18-25	867 non-null	int64
13	Totals.Alcohol.Use Past Month.26+	867 non-null	int64
14	Rates.Alcohol.Use Past Month.12-17	867 non-null	float64
15	Rates.Alcohol.Use Past Month.18-25	867 non-null	float64
16	Rates.Alcohol.Use Past Month.26+	867 non-null	float64
17	Totals.Tobacco.Cigarette Past Month.12-17	867 non-null	int64
18	Totals.Tobacco.Cigarette Past Month.18-25	867 non-null	int64
19	Totals.Tobacco.Cigarette Past Month.26+	867 non-null	int64
20	Rates.Tobacco.Cigarette Past Month.12-17	867 non-null	float64
21	Rates.Tobacco.Cigarette Past Month.18-25	867 non-null	float64
22	Rates.Tobacco.Cigarette Past Month.26+	867 non-null	float64
23	Totals.Illicit Drugs.Cocaine Used Past Year.12-17	867 non-null	int64
24	Totals.Illicit Drugs.Cocaine Used Past Year.18-25	867 non-null	int64
25	Totals.Illicit Drugs.Cocaine Used Past Year.26+	867 non-null	int64
26	Rates.Illicit Drugs.Cocaine Used Past Year.12-17	867 non-null	float64
27	Rates.Illicit Drugs.Cocaine Used Past Year.18-25	867 non-null	float64
28	Rates.Illicit Drugs.Cocaine Used Past Year.26+	867 non-null	float64
29	Totals.Marijuana.New Users.12-17	867 non-null	int64
30	Totals.Marijuana.New Users.18-25	867 non-null	int64
31	Totals.Marijuana.New Users.26+	867 non-null	int64
32	Rates.Marijuana.New Users.12-17	867 non-null	float64
33	Rates.Marijuana.New Users.18-25	867 non-null	float64
34	Rates.Marijuana.New Users.26+	867 non-null	float64
35	Totals.Marijuana.Used Past Month.12-17	867 non-null	int64
36	Totals.Marijuana.Used Past Month.18-25	867 non-null	int64
37	Totals.Marijuana.Used Past Month.26+	867 non-null	int64
38	Rates.Marijuana.Used Past Month.12-17	867 non-null	float64
39	Rates.Marijuana.Used Past Month.18-25	867 non-null	float64
40	Rates.Marijuana.Used Past Month.26+	867 non-null	float64
41	Totals.Marijuana.Used Past Year.12-17	867 non-null	int64
42	Totals.Marijuana.Used Past Year.18-25	867 non-null	int64
43	Totals.Marijuana.Used Past Year.26+	867 non-null	int64
44	Rates.Marijuana.Used Past Year.12-17	867 non-null	float64
45	Rates.Marijuana.Used Past Year.18-25	867 non-null	float64
46	Rates.Marijuana.Used Past Year.26+	867 non-null	float64
47	Totals.Tobacco.Use Past Month.12-17	867 non-null	int64
48	Totals.Tobacco.Use Past Month.18-25	867 non-null	int64
49	Totals.Tobacco.Use Past Month.26+	867 non-null	int64
50	Rates.Tobacco.Use Past Month.12-17	867 non-null	float64
51	Rates.Tobacco.Use Past Month.18-25	867 non-null	float64
52	Rates.Tobacco.Use Past Month.26+	867 non-null	float64

```
dtypes: float64(24), int64(28), object(1)
```

```
memory usage: 359.1+ KB
```

```
In [8]: df.shape
```

```
Out[8]: (867, 53)
```

```
In [9]: df.isnull().sum()
```

```

Out[9]: State                                0
        Year                                0
        Population.12-17                    0
        Population.18-25                    0
        Population.26+                      0
        Totals.Alcohol.Use Disorder Past Year.12-17  0
        Totals.Alcohol.Use Disorder Past Year.18-25  0
        Totals.Alcohol.Use Disorder Past Year.26+    0
        Rates.Alcohol.Use Disorder Past Year.12-17   0
        Rates.Alcohol.Use Disorder Past Year.18-25   0
        Rates.Alcohol.Use Disorder Past Year.26+     0
        Totals.Alcohol.Use Past Month.12-17          0
        Totals.Alcohol.Use Past Month.18-25          0
        Totals.Alcohol.Use Past Month.26+            0
        Rates.Alcohol.Use Past Month.12-17           0
        Rates.Alcohol.Use Past Month.18-25           0
        Rates.Alcohol.Use Past Month.26+             0
        Totals.Tobacco.Cigarette Past Month.12-17    0
        Totals.Tobacco.Cigarette Past Month.18-25    0
        Totals.Tobacco.Cigarette Past Month.26+      0
        Rates.Tobacco.Cigarette Past Month.12-17     0
        Rates.Tobacco.Cigarette Past Month.18-25     0
        Rates.Tobacco.Cigarette Past Month.26+       0
        Totals.Illicit Drugs.Cocaine Used Past Year.12-17  0
        Totals.Illicit Drugs.Cocaine Used Past Year.18-25  0
        Totals.Illicit Drugs.Cocaine Used Past Year.26+    0
        Rates.Illicit Drugs.Cocaine Used Past Year.12-17   0
        Rates.Illicit Drugs.Cocaine Used Past Year.18-25   0
        Rates.Illicit Drugs.Cocaine Used Past Year.26+     0
        Totals.Marijuana.New Users.12-17             0
        Totals.Marijuana.New Users.18-25             0
        Totals.Marijuana.New Users.26+              0
        Rates.Marijuana.New Users.12-17             0
        Rates.Marijuana.New Users.18-25             0
        Rates.Marijuana.New Users.26+              0
        Totals.Marijuana.Used Past Month.12-17        0
        Totals.Marijuana.Used Past Month.18-25        0
        Totals.Marijuana.Used Past Month.26+          0
        Rates.Marijuana.Used Past Month.12-17         0
        Rates.Marijuana.Used Past Month.18-25         0
        Rates.Marijuana.Used Past Month.26+           0
        Totals.Marijuana.Used Past Year.12-17         0
        Totals.Marijuana.Used Past Year.18-25         0
        Totals.Marijuana.Used Past Year.26+          0
        Rates.Marijuana.Used Past Year.12-17         0
        Rates.Marijuana.Used Past Year.18-25         0
        Rates.Marijuana.Used Past Year.26+           0
        Totals.Tobacco.Use Past Month.12-17          0
        Totals.Tobacco.Use Past Month.18-25          0
        Totals.Tobacco.Use Past Month.26+            0
        Rates.Tobacco.Use Past Month.12-17           0
        Rates.Tobacco.Use Past Month.18-25           0
        Rates.Tobacco.Use Past Month.26+             0
        dtype: int64

```

```
In [10]: df.duplicated().sum()
```

```
Out[10]: np.int64(0)
```

In [11]:

df.describe()

Out[11]:

	Year	Population.12-17	Population.18-25	Population.26+	Totals.Alcohol.Use Disorder Past Year.12-17
count	867.000000	8.670000e+02	8.670000e+02	8.670000e+02	867.000000
mean	2010.000000	4.897141e+05	6.588800e+05	3.874155e+06	19.224913
std	4.901807	5.637959e+05	7.559898e+05	4.320776e+06	25.290532
min	2002.000000	3.055100e+04	5.739500e+04	3.101100e+05	0.000000
25%	2006.000000	1.315405e+05	1.742935e+05	1.027871e+06	5.000000
50%	2010.000000	3.396850e+05	4.562400e+05	2.698757e+06	11.000000
75%	2014.000000	5.410950e+05	7.468080e+05	4.509094e+06	24.000000
max	2018.000000	3.293484e+06	4.469106e+06	2.591772e+07	204.000000

8 rows × 52 columns

In [12]:

df.dtypes

```

Out[12]: State                                     object
Year                                              int64
Population.12-17                                  int64
Population.18-25                                  int64
Population.26+                                    int64
Totals.Alcohol.Use Disorder Past Year.12-17      int64
Totals.Alcohol.Use Disorder Past Year.18-25      int64
Totals.Alcohol.Use Disorder Past Year.26+        int64
Rates.Alcohol.Use Disorder Past Year.12-17       float64
Rates.Alcohol.Use Disorder Past Year.18-25       float64
Rates.Alcohol.Use Disorder Past Year.26+         float64
Totals.Alcohol.Use Past Month.12-17              int64
Totals.Alcohol.Use Past Month.18-25              int64
Totals.Alcohol.Use Past Month.26+                int64
Rates.Alcohol.Use Past Month.12-17               float64
Rates.Alcohol.Use Past Month.18-25               float64
Rates.Alcohol.Use Past Month.26+                 float64
Totals.Tobacco.Cigarette Past Month.12-17        int64
Totals.Tobacco.Cigarette Past Month.18-25        int64
Totals.Tobacco.Cigarette Past Month.26+          int64
Rates.Tobacco.Cigarette Past Month.12-17         float64
Rates.Tobacco.Cigarette Past Month.18-25         float64
Rates.Tobacco.Cigarette Past Month.26+           float64
Totals.Illicit Drugs.Cocaine Used Past Year.12-17 int64
Totals.Illicit Drugs.Cocaine Used Past Year.18-25 int64
Totals.Illicit Drugs.Cocaine Used Past Year.26+   int64
Rates.Illicit Drugs.Cocaine Used Past Year.12-17 float64
Rates.Illicit Drugs.Cocaine Used Past Year.18-25 float64
Rates.Illicit Drugs.Cocaine Used Past Year.26+    float64
Totals.Marijuana.New Users.12-17                 int64
Totals.Marijuana.New Users.18-25                 int64
Totals.Marijuana.New Users.26+                   int64
Rates.Marijuana.New Users.12-17                  float64
Rates.Marijuana.New Users.18-25                  float64
Rates.Marijuana.New Users.26+                    float64
Totals.Marijuana.Used Past Month.12-17           int64
Totals.Marijuana.Used Past Month.18-25           int64
Totals.Marijuana.Used Past Month.26+             int64
Rates.Marijuana.Used Past Month.12-17            float64
Rates.Marijuana.Used Past Month.18-25            float64
Rates.Marijuana.Used Past Month.26+              float64
Totals.Marijuana.Used Past Year.12-17            int64
Totals.Marijuana.Used Past Year.18-25            int64
Totals.Marijuana.Used Past Year.26+              int64
Rates.Marijuana.Used Past Year.12-17             float64
Rates.Marijuana.Used Past Year.18-25             float64
Rates.Marijuana.Used Past Year.26+               float64
Totals.Tobacco.Use Past Month.12-17              int64
Totals.Tobacco.Use Past Month.18-25              int64
Totals.Tobacco.Use Past Month.26+                int64
Rates.Tobacco.Use Past Month.12-17               float64
Rates.Tobacco.Use Past Month.18-25               float64
Rates.Tobacco.Use Past Month.26+                 float64
dtype: object

```

```

In [13]: print (df["State"])

```

```

0         Alabama
1         Alaska
2         Arizona
3         Arkansas
4         California
...
862        Virginia
863        Washington
864        West Virginia
865        Wisconsin
866        Wyoming
Name: State, Length: 867, dtype: object

```

```
In [14]: df["State"] = df["State"].str.strip()
```

```
In [15]: df["Year"] = pd.to_numeric(df["Year"], errors="coerce")
```

```
In [16]: df = df.drop_duplicates()
```

```
In [17]: print("Dataset shape after cleaning:", df.shape)
```

Dataset shape after cleaning: (867, 53)

The data shape remains the same because our data set has no missing values or duplicate values.

```
In [26]: df_transformed.shape
```

```
Out[26]: (867, 103)
```

Feature Engineering

Two new features are created for this analysis. Firstly, an average substance-use rate is calculated from alcohol use disorder, cocaine use, and tobacco use among individuals aged 18–25. This feature provides a combined measure of other substance-use patterns that may be associated with marijuana use.

Secondly, a young-adult population ratio is created by dividing the population aged 18–25 by the total population represented by the three age groups (12–17, 18–25, and 26+). This transforms the population counts into a proportion and provides information about the relative size of the 18–25 age group.

```
In [50]: df["Average_Other_Substance_Use_Rate_18_25"] = df[
    [
        "Rates.Alcohol.Use Disorder Past Year.18-25",
        "Rates.Illicit Drugs.Cocaine Used Past Year.18-25",
        "Rates.Tobacco.Use Past Month.18-25"
    ]
].mean(axis=1)
```

```
In [52]: df["Young_Adult_Population_Ratio"] = (
    df["Population.18-25"] /
    (
        df["Population.12-17"] +
        df["Population.18-25"] +
        df["Population.26+"]
    )
)
```

```
)
)
```

```
In [53]: df[
    [
        "Rates.Alcohol.Use Disorder Past Year.18-25",
        "Rates.Illicit Drugs.Cocaine Used Past Year.18-25",
        "Rates.Tobacco.Use Past Month.18-25",
        "Average_Other_Substance_Use_Rate_18_25",
        "Young_Adult_Population_Ratio"
    ]
].head()
```

```
Out[53]:
```

	Rates.Alcohol.Use Disorder Past Year.18-25	Rates.Illicit Drugs.Cocaine Used Past Year.18-25	Rates.Tobacco.Use Past Month.18-25	Average_Other_Substance_Use_R
0	136.490	54.892	451.976	2
1	187.891	83.680	484.270	2
2	193.626	85.300	397.968	2
3	175.913	60.500	558.846	2
4	148.312	66.052	351.353	1

Define Features and Target

The target variable for this machine learning task is the rate of marijuana use in the past year among individuals aged 18–25. The selected features include state, year, young adult population ratio, alcohol use disorder, cocaine use, and tobacco use among the same age group.

```
In [54]: features = [
    "State",
    "Year",
    "Young_Adult_Population_Ratio",
    "Rates.Alcohol.Use Disorder Past Year.18-25",
    "Rates.Illicit Drugs.Cocaine Used Past Year.18-25",
    "Rates.Tobacco.Use Past Month.18-25"
]

X = df[features]
y = df["Rates.Marijuana.Used Past Year.18-25"]
```

```
In [55]: print("Features:")
print(X.head())

print("\nTarget:")
print(y.head())
```


Features:

	State	Year	Young_Adult_Population_Ratio	\
0	Alabama	2002	0.135237	
1	Alaska	2002	0.125419	
2	Arizona	2002	0.136343	
3	Arkansas	2002	0.135905	
4	California	2002	0.137757	

	Rates.Alcohol.Use Disorder Past Year.18-25	\
0	136.490	
1	187.891	
2	193.626	
3	175.913	
4	148.312	

	Rates.Illicit Drugs.Cocaine Used Past Year.18-25	\
0	54.892	
1	83.680	
2	85.300	
3	60.500	
4	66.052	

	Rates.Tobacco.Use Past Month.18-25
0	451.976
1	484.270
2	397.968
3	558.846
4	351.353

Target:

0	237.880
1	389.026
2	275.435
3	288.856
4	282.887

Name: Rates.Marijuana.Used Past Year.18-25, dtype: float64

Train/Test Split The dataset is divided into training and testing sets using an 80/20 split respectively. The training data will be used to learn the preprocessing parameters and train the machine learning model, while the test data will be reserved for final evaluation. A random state of 42 is used to ensure that the same split can be reproduced.

```
In [56]: X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.20, random_state=42)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

Training data: (693, 6)

Testing data: (174, 6)

Identify Feature Types The selected features are divided into numerical and categorical variables. Numerical features include year, young adult population, alcohol use disorder, cocaine use, and tobacco use. These variables require numerical preprocessing. State is treated as a categorical feature because it represents different state categories rather than a continuous numerical measurement. It will be encoded using one-hot encoding.

```
In [59]: numeric_features = [
        "Year",
```

```

    "Young_Adult_Population_Ratio",
    "Rates.Alcohol.Use Disorder Past Year.18-25",
    "Rates.Illicit Drugs.Cocaine Used Past Year.18-25",
    "Rates.Tobacco.Use Past Month.18-25"
]

categorical_features = ["State"]

```

Numerical Feature Preprocessing

The numerical features are processed using a two-step pipeline. The missing numerical values are replaced with the median value (it is less sensitive to outliers) of the corresponding feature, then `StandardScaler` is used to standardize the numerical variables so that they are placed on a comparable scale.

The preprocessing steps are fitted using the training data to avoid data leakage.

```

In [60]: numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

```

Categorical Feature Preprocessing

The State variable is categorical, so `OneHotEncoder` is used to transform each state into numerical binary features. The missing values are replaced with the most frequent state and the `'handle_unknown="ignore"'` option ensures that the preprocessing pipeline does not produce an error if a category appears in the test data that was not present in the training data.

```

In [61]: categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])

```

Combine Preprocessing Steps

`ColumnTransformer` is used to apply the appropriate preprocessing method to each type of feature. Numerical variables are passed through the numerical preprocessing pipeline, and the State variable is passed through the categorical pipeline, where it is one-hot encoded.

```

In [62]: preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_transformer, numeric_features),
        ("cat", categorical_transformer, categorical_features)
    ]
)

```

```

In [63]: X_train_processed = preprocessor.fit_transform(X_train)

print("Original Training Data Shape:", X_train.shape)
print("Processed Training Data Shape:", X_train_processed.shape)

```

Original Training Data Shape: (693, 6)

Processed Training Data Shape: (693, 56)

```
In [64]: X_test_processed = preprocessor.transform(X_test)

print("Original Testing Data Shape:", X_test.shape)
print("Processed Testing Data Shape:", X_test_processed.shape)
```

Original Testing Data Shape: (174, 6)

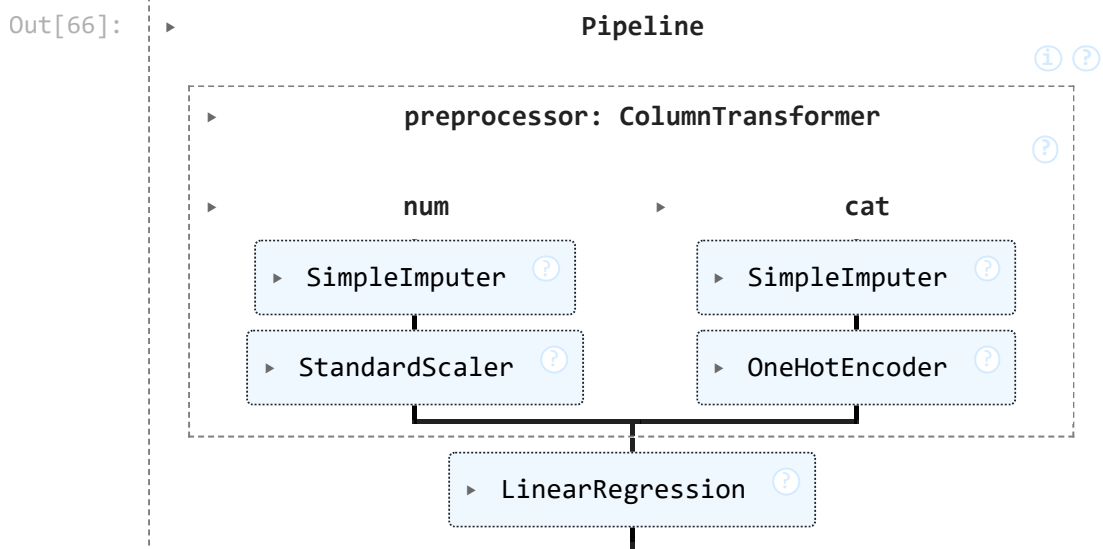
Processed Testing Data Shape: (174, 56)

The machine learning pipeline is trained using the training dataset. The pipeline contains both the preprocessing steps and the Linear Regression model. Using the pipeline ensures that the preprocessing operations are fitted only on the training data. This helps prevent data leakage because information from the test dataset is not used during the training process.

```
In [65]: from sklearn.linear_model import LinearRegression

model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", LinearRegression())
])
```

```
In [66]: model.fit(X_train, y_train)
```



A 5-fold cross-validation is performed on the training dataset, the training data is divided into five subsets, and the model is trained on four folds and validated on the remaining fold. This process is repeated five times so that each fold is used as the validation set once.

This helps determine whether the model performs consistently across different subsets of the training data and reduces the possibility of relying on a single train-validation split. The test dataset remains untouched during cross-validation and is reserved for the final evaluation of the trained model.

```
In [69]: from sklearn.model_selection import cross_val_score
```

```

cv_scores = cross_val_score(
    model,
    X_train,
    y_train,
    cv=5,
    scoring="r2"
)

print("Cross-validation scores:", cv_scores)
print("Mean CV score:", cv_scores.mean())

```

Cross-validation scores: [0.87238051 0.91008496 0.89316609 0.90065781 0.84554001]
Mean CV score: 0.8843658757785974

In [70]: `y_pred = model.predict(X_test)`

In [71]: `results = pd.DataFrame({`
 `"Actual": y_test.values,`
 `"Predicted": y_pred`
`})`
`results.head()`

Out[71]:

	Actual	Predicted
0	328.703	319.991365
1	283.836	281.011707
2	239.584	241.165901
3	357.991	339.463242
4	388.969	380.564676

Model Evaluation The trained model is evaluated using the test dataset. The evaluation metrics used are Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

In [72]: `from sklearn.metrics import (`
 `mean_absolute_error,`
 `mean_squared_error,`
 `r2_score`
`)`
`mae = mean_absolute_error(y_test, y_pred)`
`mse = mean_squared_error(y_test, y_pred)`
`rmse = mse ** 0.5`
`r2 = r2_score(y_test, y_pred)`
`print("MAE:", mae)`
`print("MSE:", mse)`
`print("RMSE:", rmse)`
`print("R²:", r2)`

MAE: 17.76711940014197
MSE: 566.0173470509997
RMSE: 23.791119079417
 R^2 : 0.8703477286774512

```
In [73]: print("Mean CV R²:", cv_scores.mean())  
         print("Test R²:", r2)
```

Mean CV R²: 0.8843658757785974

Test R²: 0.8703477286774512

Overall, the model demonstrates a relatively strong predictive performance based on the R^2 score. However, the MAE and RMSE indicate that some prediction errors remain. The model performed similarly during cross-validation and on the test set. The small difference suggests that the model's performance is fairly consistent and that there isn't a major indication of overfitting.